UNIVERSITY OF OSLO

A Self-Adaptive Digital Twin Architecture to Automate Greenhouse Management

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Rationale

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- They need to adapt to changing conditions and ensure they can capture the correct states over time
- Self-adaptation can improve reliability and reduce manual intervention

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  - Adaptive control strategies for watering under changing conditions
- Validation in both sandbox and physical greenhouse environments

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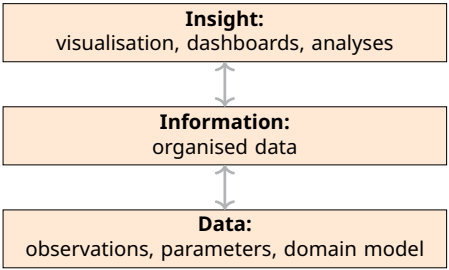
Digital Twins components



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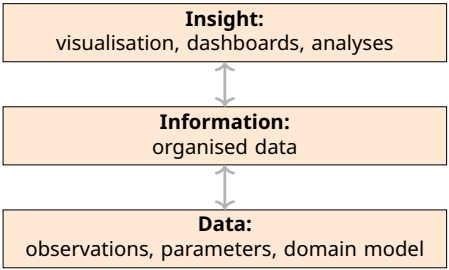

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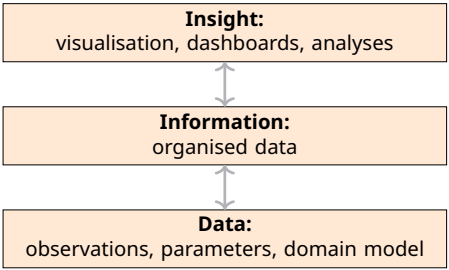

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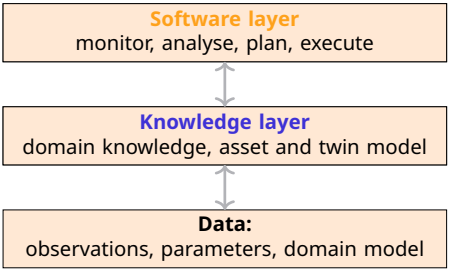

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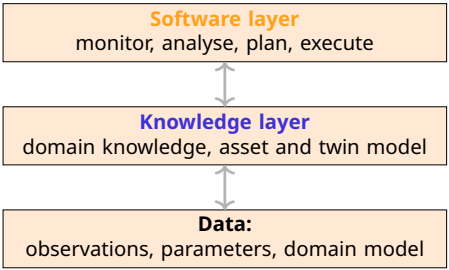

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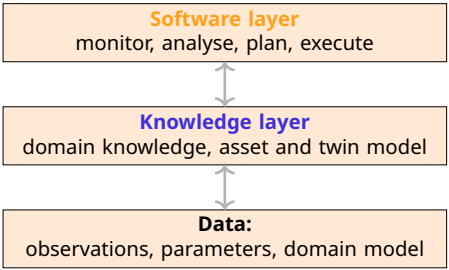

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- States of the DT cycles over time and they need to be captured

Twinning in SMOL

Programming support for twins with a behavioural and a structural layer

smolang.org

Twinning in SMOL

Programming support for twins with a behavioural and a structural layer

SMOL: Semantic Model Object Language

- Smalll OO programming system
- Ontology reasoners allow querying the KB
- Used to create the digital model

smolang.org

Twinning in SMOL

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behavioural twins in SMOL

- SMOL can encapsulate (simulation) models based on the FMI standard
- Can automatically adapt the object states in the KB at runtime through semantic lifting

Semantic Lifting

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- It allows the DT to capture the current state of the physical system and reason about it
- This enables dynamic reclassification of entities and supports adaptation based on the latest observations
- By lifting static data into a semantic model, the DT can maintain an up-to-date representation of the physical system's conditions and make informed decisions for control and adaptation

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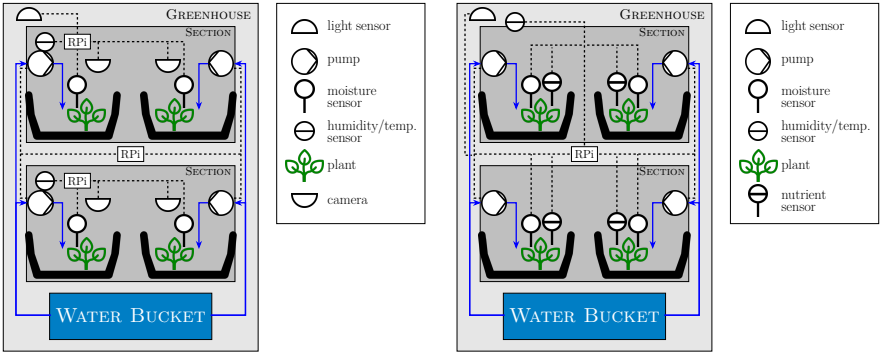
- Query inferred knowledge (SPARQL) to select adaptation and control actions

Two Greenhouse Setups



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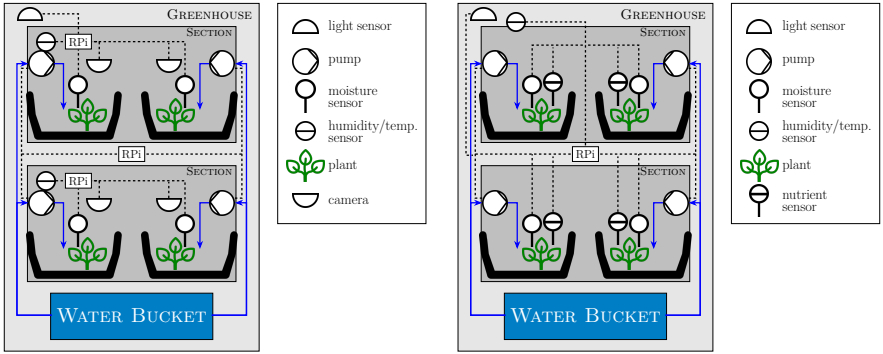

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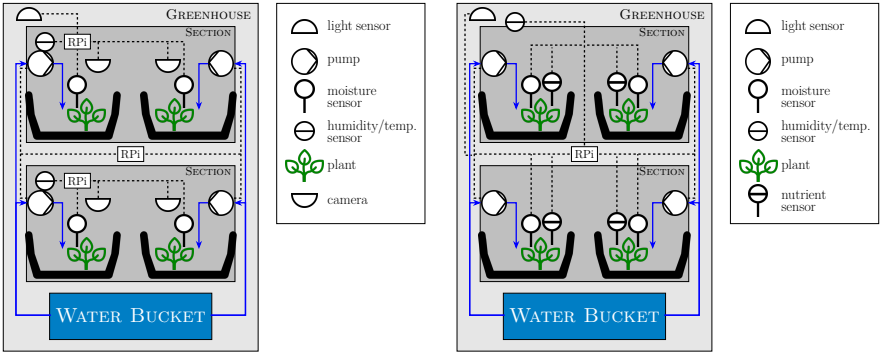
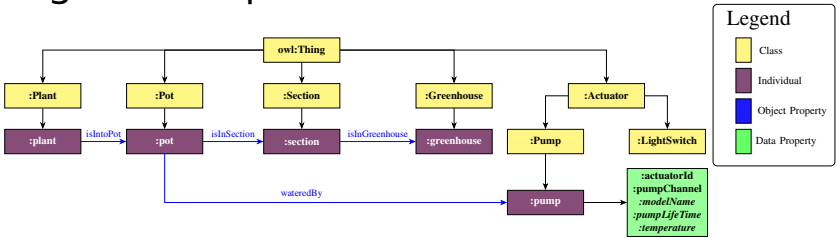


- Two different physical setups with different sensors, actuators, and plant types
- Both setups share the same DT logic through a generic ontology model

Knowledge Base Representation



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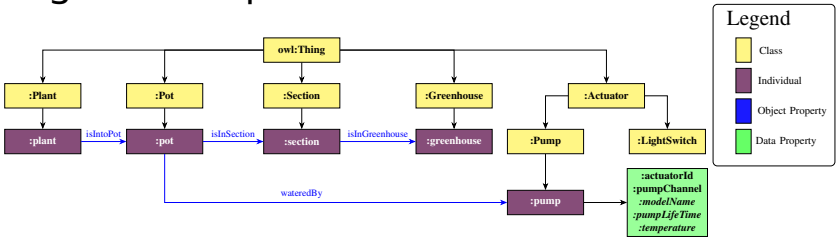

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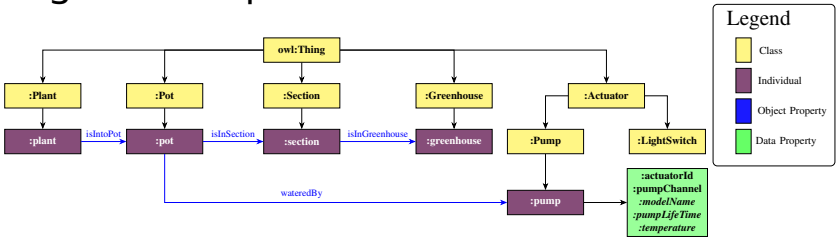

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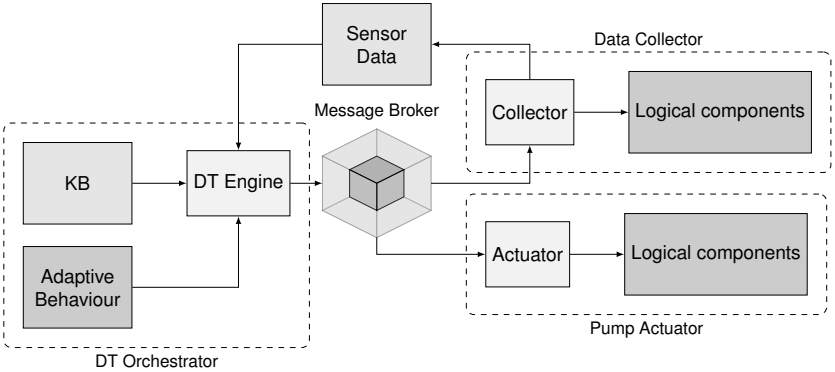
- A common OWL-based model captures plants, sensors, actuators, and environment

Architecture Overview



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- **Pump Actuator**: executes watering actions produced by selected strategies
- **Message Broker**: decouples communication across distributed components

Declarative Adaptation

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- Based on the state of the plant, the appropriate water strategy is applied

Declarative Adaptation (cont.)

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- This approach allows for a clear separation of concerns, where the adaptation logic is defined declaratively in the ontology, and the DT Orchestrator simply executes the reasoning results
- Compared to hardcoding the adaptation logic in the software, this ontology-based approach provides greater flexibility and maintainability, as changes to the adaptation rules can be made directly in the ontology without modifying the underlying code

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Takaways

- The proposed architecture enables reliable self-adaptive management of physical systems through a digital twin

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- The proposed architecture enables reliable self-adaptive management of physical systems through a digital twin
- Dynamic adaptation and strategy adaptation worked in sandbox and physical settings
- The generic model supports multiple physical systems with heterogeneous configurations

Future Work

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- Improve generic KB modelling for more diverse physical-twin categories
- Reduce semantic lifting overhead for real-time deployment scenarios
- Learn from past experience to improve the values for watering or the period to water

Thanks for the attention

Any questions?

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